

Green Chemistry

- Green chemistry is a way of thinking and is about utilising the existing knowledge and principles of chemistry and other sciences to reduce the adverse impact on environment.
- Utilisation of existing knowledge base for reducing the chemical hazards along with the developmental activities is the foundation of green chemistry.
- Green chemistry , is a cost effective approach which involves reduction in material, energy consumption and waste generation

Green Chemistry in day-to-day Life

(i) Dry Cleaning of Clothes

- Tetra chloroethene ($\text{Cl}_2\text{C}=\text{CCl}_2$) was earlier used as solvent for dry cleaning.
- The compound contaminates the ground water and is also a suspected carcinogen
- The process using this compound is now being replaced by a process, where liquefied carbondioxide, with a suitable detergent is used.
- Replacement of halogenated solvent by liquid CO_2 will result in less harm to ground water.

(ii) Bleaching of Paper

- Chlorine gas was used earlier for bleaching paper. These days, hydrogen peroxide (H_2O_2) with suitable catalyst, which promotes the bleaching action of hydrogen peroxide, is used.
- hydrogen peroxide (H_2O_2) is used for the purpose of bleaching clothes in the process of laundry, which gives better results and makes use of lesser amount of water

Strategies for controlling environmental pollution can be:

- (i) waste management i.e., reduction of the waste and proper disposal, also recycling of materials and energy,
- (ii) adopting methods in day-to-day life, which results in the reduction of environmental pollution.